



Comsats University Islamabad, Lahore Campus

Assignment <2> – SPRING 2025

Course Title	Information Security	Course Code	CSC432	Credit Hours	3(2,1)
Instructor/s	Dr. Muhammad Sharjeel	Program Name	BSSE		
Semester	8 th	Batch	FA21	Section	A
Time Allowed	1 week	Maximum Marks			
Student Name:	Aoun-Haider	Reg No.	FA21-BSE-133		

Question Statement

[CLO-2]; Bloom Taxonomy Level: <Applying>

Research ARP spoofing and ARP poisoning. Write a paragraph explaining your understanding of how ARP spoofing relates to data interception, session hijacking, and denial-of-service attacks. Download and install Ettercap. Demonstrate a MITM attack with Ettercap using ARP spoofing. Read about Cain & Abel (software) and explain how it relates to Ettercap. Describe how a simple command 'arp -a' provides protection against ARP poisoning. What are the other ways to prevent ARP poisoning.

Solution:

ARP (Address Resolution Protocol) is used to map IP address to MAC address in a local network. For example, if I want to communicate with some other device, my device has to figure out its MAC address using its IP address. My device will broadcast a packet in local network containing IP address of other device, and the corresponding device to which this IP belongs will respond.

My Device: Who has IP 192.168.0.1? Tell me your MAC address

Other Device(s): **Ignore the message**

Target Device: I have IP 192.168.0.1 and MAC address is AA: BB: CC: DD: EE: FF

Intruder: **Hijacks the message** [message: I have IP 192.168.0.1 and MAC address is BD: FA: CC: DD: BB: AA]

My Device: I am sending you 3000 rupees, tell me your account no.

What is ARP Spoofing and ARP Poisoning?

Intruder pretends to have the IP address of the other device and returns his MAC address (ARP Spoofing) and now receives the traffic from my device. Intruder can also corrupt the ARP table to disable communication with other devices. So, he can send a fake reply or can get access to ARP table and corrupt it or modify a specific entry against IP address.

ARP Spoofing focus is faking identity while *ARP Poisoning* can alter ARP cache entries or corrupt it to disrupt/intercept communication.

ARP Spoofing = Pretending to be another device

ARP Poisoning = Corrupting the ARP cache of a victim

How ARP spoofing relates to data interception, session hijacking, and denial-of-service attacks?

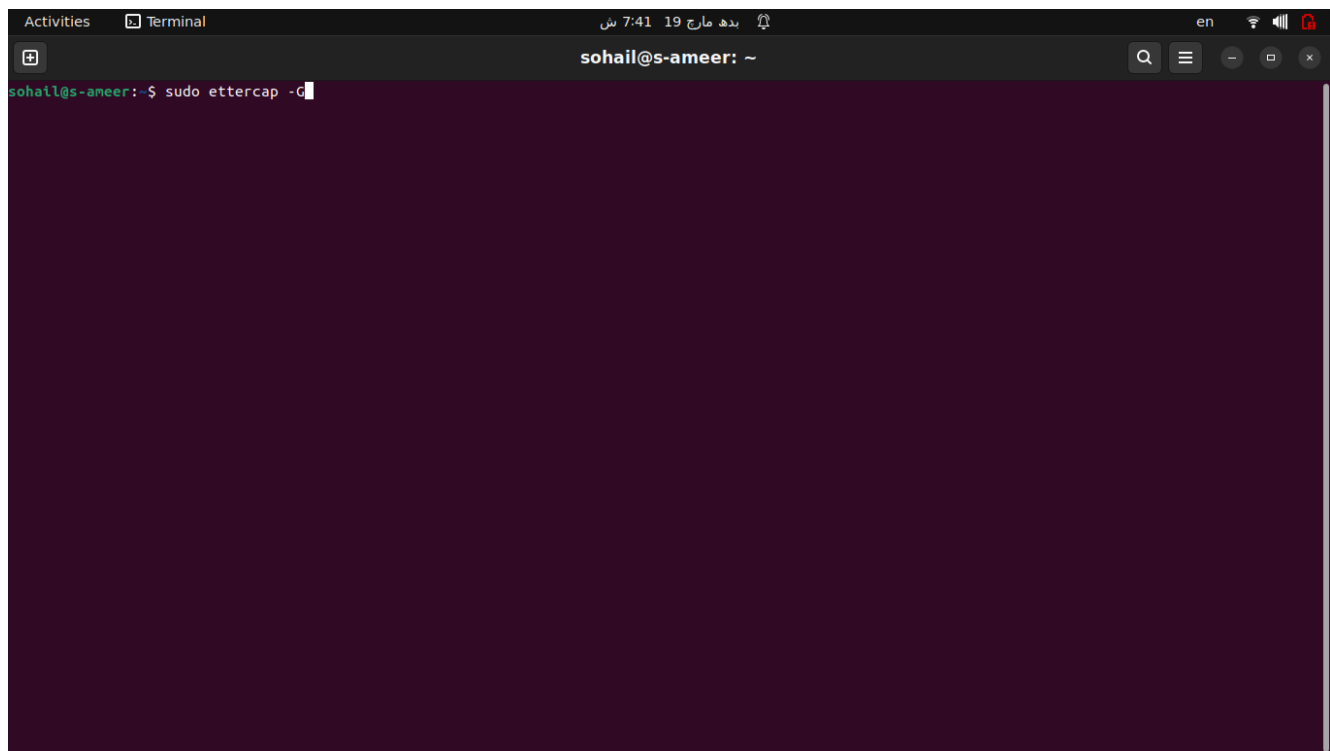
The attacker spoofs the MAC address of the victim using ARP Spoofing and can now read, modify and forward the data to the actual router so the victim remains unaware. (Data Interception, Man in the middle attack MITM)

The attacker takes over a user's active session like login session for a bank or social media account. Attacker performs ARP spoofing to intercept traffic and then steals the victim's cookie session and finally reuses the session cookie to impersonate the victim. (Session Hijacking)

Attacker can disrupt the network communication making it unusable for victims. The attacker poises ARP tables of multiple devices causing them to send packets to the wrong destination. Victim's device cannot communicate properly leading to network failure or slow performance. (Denial of Service Dos)

Demonstrate a MITM attack with Ettercap using ARP spoofing.

Starting Ettercap:



Starting Ettercap (Cont):

Activities

Ettercap

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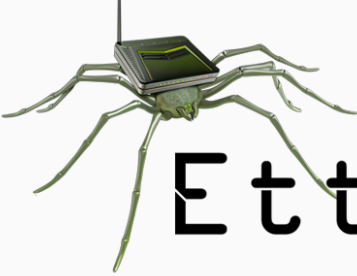
en

sohail@s-ameer: ~

Unpacking liblua5.1-common (2.1.0-beta3+dfsg-6) ...
Selecting previously unselected package liblua5.1-2:amd64.
Preparing to unpack .../4-liblua5.1-2_2.1.0-beta3+dfsg-6_amd64.deb ...
Unpacking liblua5.1-2:amd64 (2.1.0-beta3+dfsg-6) ...
Selecting previously unselected package libnet1:amd64 (1.1.6+dfsg-3.1build3) ...
Preparing to unpack .../5-libnet1_1.1.6+dfsg-3.1build3_amd64.deb ...
Unpacking libnet1:amd64 (1.1.6+dfsg-3.1build3) ...
Selecting previously unselected package libgeoip1:amd64 (1.5.16-1ubuntu0.1) ...
Preparing to unpack .../6-libgeoip1_1.5.16-1ubuntu0.1_amd64.deb ...
Unpacking libgeoip1:amd64 (1.5.16-1ubuntu0.1) ...
Selecting previously unselected package liblua5.1-2:amd64 (2.1.0-beta3+dfsg-6) ...
Preparing to unpack .../7-liblua5.1-2_2.1.0-beta3+dfsg-6_amd64.deb ...
Unpacking liblua5.1-2:amd64 (2.1.0-beta3+dfsg-6) ...
Setting up libnet1:amd64 (1.1.6+dfsg-3.1build3) ...
Setting up liblua5.1-common (2.1.0-beta3+dfsg-6) ...
Setting up libgeoip1:amd64 (1.5.16-1ubuntu0.1) ...
Setting up geoip-database (20220927-1) ...
Setting up ethtool (1:5.16-1ubuntu0.1) ...
Setting up liblua5.1-2:amd64 (2.1.0-beta3+dfsg-6) ...
Setting up ettercap-common (0.8.3.1-5) ...
Setting up ettercap-graphical (0.8.3.1-5) ...
Processing triggers for man-db (2.9.4-2) ...
Processing triggers for mailcap (3.69) ...
Processing triggers for desktop-file-utils (0.23-1ubuntu3) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for libc-bin (2.34-0ubuntu2) ...
sohail@s-ameer:~\$ ettercap -G
Starting Unified sniffing...
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!
ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team
ettercap 0.8.3.1
sohail@s-ameer:~\$ sudo ettercap -G
ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team
Failed to create secure directory (/root/.config/pulse): Permission denied
Failed to create secure directory (/root/.config/pulse): Permission denied

Ettercap

0.8.3.1 (EB)



Ettercap

Adding target host:

Activities

Ettercap

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sohail@s-ameer: ~

Unpacking libnet1:amd64 (1.1.6+dfsg-3.1build3) ...
Selecting previously unselected package ettercap-common.
Preparing to unpack .../6-ettercap-common_1%3a0.8.3.1-5_amd64.deb ...
Unpacking ettercap-common (0.8.3.1-5) ...
Selecting previously unselected package libnet1:amd64 (1.1.6+dfsg-3.1build3) ...
Preparing to unpack .../7-libnet1_1.1.6+dfsg-3.1build3_amd64.deb ...
Unpacking libnet1:amd64 (1.1.6+dfsg-3.1build3) ...
Setting up libnet1:amd64 (1.1.6+dfsg-3.1build3) ...
Setting up liblua5.1-common (2.1.0-beta3+dfsg-6) ...
Setting up libgeoip1:amd64 (1.5.16-1ubuntu0.1) ...
Setting up geoip-database (20220927-1) ...
Setting up ethtool (1:5.16-1ubuntu0.1) ...
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sohail@s-ameer:~\$ ettercap -G
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Ettercap

0.8.3.1 (EB)

Host List

IP Address	MAC Address	Description
fe80::870:91ff:fecc:2dbc	0A:70:91:CC:2D:BC	
fe80::10e3:a956:ae16:2c00	A2:E3:76:AB:5A:78	
fe80::6283:e7ff:fe81:ae10	60:83:E7:81:AE:10	
fe80::64bd:3aff:fe1e:9f4d	66:BD:3A:1E:9F:4D	Android-59.local
192.168.0.1	60:83:E7:81:AE:10	
192.168.0.55	66:BD:3A:1E:9F:4D	Android-59.local
fe80::84ec:deff:fe0a:a478	86:EC:DE:0A:A4:78	

Delete Host

Add to Target 1

Add to Target 2

Checking connections:

Activities

Ettercap

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en

Ettercap

0.8.3.1 (EB)

Host List

Connections

Host filter

Protocol filter

Connection state filter

☐ TCP

☒ UDP

☒ Other

☒ Active

☒ Idle

☒ Closing

☒ Closed

☒ Killed

View Details

Kill Connection

Expunge Connections

Lua: no scripts were specified, not starting up!

Starting Unified sniffing...

Host fe80::870:91ff:fecc:2dbc added to TARGET1

Host fe80::10e3:a956:ae16:2c00 added to TARGET2

Checking Profiles:

Activities

Ettercap

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Ettercap

0.8.3.1 (EB)

Host List

Connections

Profiles

IP Address	Hostname	Country
4.224.95.102	pub-csm-ince-01-t.trouter.skype.com	United States
8.8.8.8		United States
52.200.127.40	win-extension.femetrics.grammarly.io	United States
57.144.148.3	z-m-gateway.facebook.com	Belgium
57.144.148.128	connect.facebook.net	Belgium
57.144.148.141	graph.facebook.com	Belgium
57.144.148.145	gateway.instagram.com	Belgium
57.144.148.192	i.instagram.com	Belgium
57.144.149.33	g.whatsapp.net	Belgium
72.255.57.18		Pakistan

Purge Local

Purge Remote

Convert to Host List

Dump to File

Lua: no scripts were specified, not starting up!

Starting Unified sniffing...

DHCP: [02:C9:73:15:82:6D] REQUEST 192.168.0.190

Host 192.168.0.1 added to TARGET1

Host fe80::c9:73ff:fe15:826d added to TARGET2

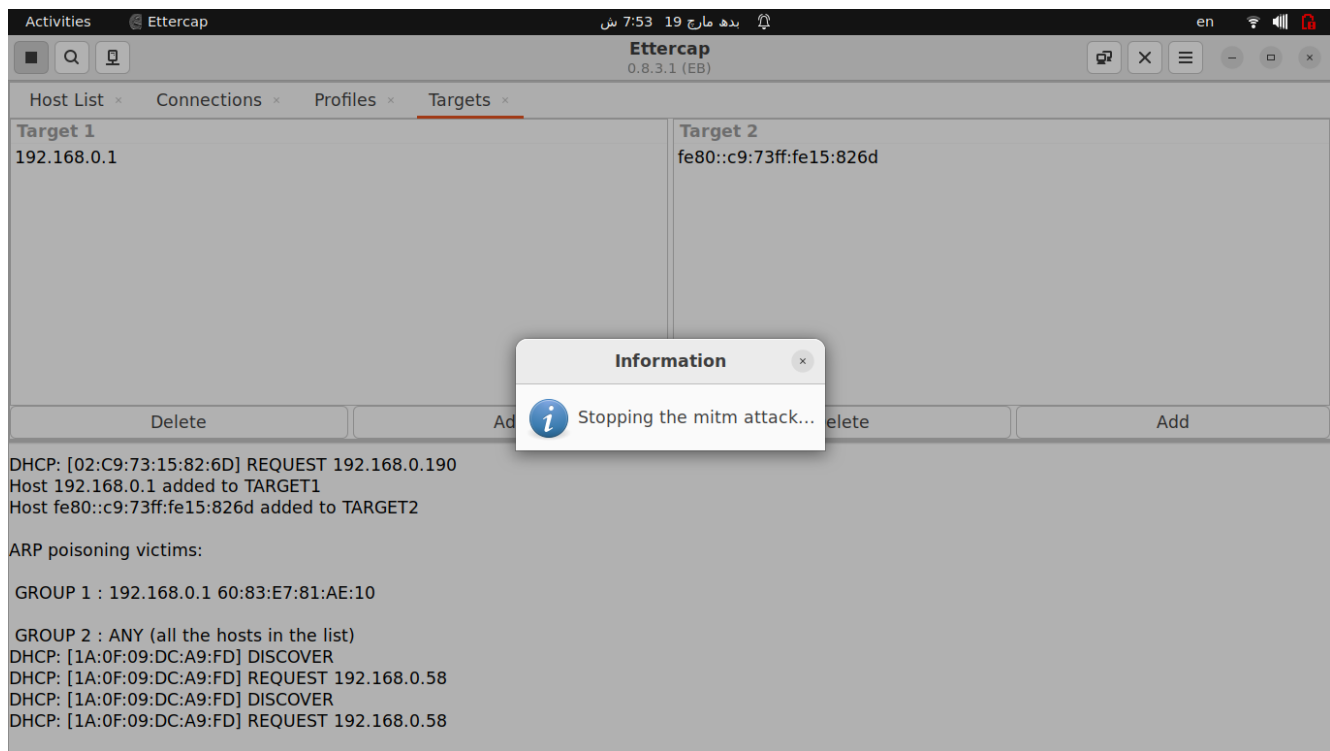
ARP poisoning victims:

GROUP 1 : 192.168.0.1 60:83:E7:81:AE:10

GROUP 2 : ANY (all the hosts in the list)

DHCP: [1A:0F:09:DC:A9:FD] DISCOVER

DHCP: [1A:0F:09:DC:A9:FD] REQUEST 192.168.0.58



How Cain & Abel (software) relates to Ettercap.

Both can be used for Man in the Middle Attack (MITM) using ARP Spoofing but differ in features, platform compatibility and attack capabilities. For advance MITM attacks, attacker could use Ettercap while for advance traffic manipulation, Cain & Abel can be used.

Describe how a simple command ‘arp -a’ provides protection against ARP poisoning.

This command shows all IP to MAC address mapping entries of ARP table. It does not provide some mechanism to protect against ARP poisoning but can help to detect it. Normally, each IP address should have a unique MAC address but if a specific MAC address exist multiple times against different IP addresses, it could indicate ARP poisoning.

What are the other ways to prevent ARP poisoning?

1. Try to manually set MAC address against IP address using “*arp -s IP_Address MAC_Address*”
2. Use network monitoring tools like Wireshark
3. If using a managed switch, enable Dynamic ARP Inspection (DAI) to filter spoofed ARP packets.
4. Write a script in some programming language which can detect multiple MAC addresses in ARP table.
5. Use VPN or secure protocols like HTTPS, SSH, TLS
6. Use IPV6 instead of IPV4 which contains built-in mechanism for secure neighbor discovery
7. Enable port security on switches